

Research-Informed Mathematics Teaching Design

Evidence-informed. Research-supported. Pilot-ready.

Kaizen Maths should be described carefully: it is not yet an independently evaluated intervention with published impact findings. It is deliberately built around well-established principles from mathematics education, learning science, formative assessment, and purposeful educational technology.

Responsible claim

Kaizen Maths gives teachers a practical workspace for applying research-informed mathematics teaching routines: structured practice, worked examples, visual representations, feedback, curriculum alignment, and teacher-led classroom use.

How the evidence connects to Kaizen Maths

1. Small-step mastery Topic levels, mixed sets, and curriculum maps help teachers sequence practice coherently.	5. Feedback Answers, worked steps, pupil tasks, mastery thresholds, and teacher review support responsive teaching.
2. Retrieval practice Fresh questions support low-stakes recall, fluency, and repeated practice without fixed sheets.	6. Curriculum coherence Coverage, textbook alignment, and school defaults reduce scattered planning and improve consistency.
3. Worked examples One Example mode and step reveal make methods visible instead of only showing answers.	7. Purposeful technology Kaizen supports teacher decisions: what to show, when to reveal it, and how to model it.
4. Representations Bar models, area models, diagrams, manipulatives, and dynamic tools expose mathematical structure.	

Virtual textbook Unlimited topic-based question generation, levels, worked examples, worksheets, assessments, and classroom displays.	Teacher-led classroom use Projection, step reveal, annotation, capture, and full-screen routines keep the teacher in control.	Assessment and feedback Pupil tasks, mastery attempts, review comments, answer keys, and worked steps turn practice into usable information.
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Next evidence step: A school or ministry pilot can test implementation and impact in a defined topic: baseline task, Kaizen-supported teaching sequence, post-task, teacher feedback, pupil work samples, and usage notes. This would move the claim from research-informed design toward local evidence of impact.

Selected research links

EEF mathematics evidence review educationendowmentfoundation.org.uk/.../mathematics-in-key-stages-2-and-3	WWC mathematics intervention guide ies.ed.gov/ncee/WWC/PracticeGuide/26/Published
WWC algebra practice guide ies.ed.gov/ncee/WWC/PracticeGuide/20	EEF feedback guidance educationendowmentfoundation.org.uk/.../guidance-reports/feedback
Dunlosky et al. learning techniques psychologicalscience.org/publications/journals/pspi/learning-techniques.html	Roediger and Karpicke testing effect doi.org/10.1111/j.1467-9280.2006.01693.x
NCETM mastery: five big ideas ncetm.org.uk/teaching-for-mastery/.../five-big-ideas-in-teaching-for-mastery/	EEF digital technology guidance educationendowmentfoundation.org.uk/.../guidance-reports/digital

Kaizen Maths - generated for school conversations, pilots, and department due diligence. Full links and context are available on the Research-Informed Design page inside the site.